

## SUMMARY TABLE: MATHEMATICS GRADE 10

### Sub-Strand 4.2: Probability I — Teacher Reference (Pre-filled)

| SUMMARY TABLE: MATHEMATICS GRADE 10 |                                                                                                                                                                                                                                     |
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| Sub-Strand                          | Sub-Strand 4.2: Probability I                                                                                                                                                                                                       |
| Driving Question                    | DRIVING QUESTION: How can we use mathematics to measure and predict the likelihood of uncertain everyday events — like Mama Njeri's mandazi sales — so we can make smarter decisions even when we cannot be certain of the outcome? |

| INSTRUCTIONS                                                                                                                                                                                                                                                                            |
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| FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson. |

| Lesson # and Title                                                                                        | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Lesson 1</b><br><b>The Mandazi Stall Mystery: Introducing Uncertainty and Experimental Probability</b> | Students observed the results of a physical simulation — rolling a die or drawing coloured tokens representing mandazi types sold at Mama Njeri's stall — recording outcomes in a frequency tally over 20–30 trials. As more trials were conducted, students noticed that the relative frequency of each outcome began to stabilise toward a consistent value. Teacher should look for students who conflate 'random' with 'unpredictable forever' — use cold-calling (minimum 3 students) to surface this misconception before the explain phase. | Students learned that (1) probability is a mathematical measure of likelihood expressed as a number between 0 and 1 inclusive (0 = impossible, 1 = certain); (2) a random experiment is any process whose outcome cannot be known in advance; (3) the sample space (S) is the complete set of all possible outcomes; (4) an event (E) is any subset of the sample space; and (5) experimental (empirical) probability is calculated as $P(E) = (\text{number of times E occurs}) \div (\text{total number of trials})$ . Key vocabulary: experiment, outcome, sample space, event, experimental probability, relative frequency. | Teacher should emphasise that 'random' does not mean 'impossible to analyse' — it means the outcome of any single trial is uncertain, but patterns emerge over many trials. Connect explicitly to the driving question: Mama Njeri cannot know exactly how many plain mandazi she will sell tomorrow, but after recording 30 days of sales data she can assign a probability. Pedagogical note: Use the Law of Large Numbers informally here — more trials → relative frequency → true probability. Do NOT introduce the formal theorem; plant the intuition. Ensure students record $P(E)$ as a fraction, decimal, and percentage for every event discussed, reinforcing numerical fluency. |
| <b>Lesson 2</b><br><b>Experimental Probability II: More Trials, More Patterns</b>                         | Students pooled class-wide trial data (combining individual tallies from Lesson 1 into one large dataset of 100+ trials) and plotted relative frequency against number of trials on a line graph. They observed that early trial results fluctuate wildly but the graph 'settles' as trials increase — a direct, visible demonstration of the Law of Large                                                                                                                                                                                         | Students formally learned that experimental probability $P(E) = (\text{number of favourable outcomes}) \div (\text{total number of trials})$ , and that this value becomes increasingly reliable as the number of trials grows. They learned that experimental probability is always an estimate — it approximates, but may not equal, the theoretical probability.                                                                                                                                                                                                                                                              | Teacher should reinforce why more data is better using the mandazi context: if Mama Njeri only records 3 days of sales, her estimates are unreliable; after 60 days, her estimates are trustworthy enough to inform how many batches to fry each morning. This directly answers the driving question — mathematics helps make smarter                                                                                                                                                                                                                                                                                                                                                        |

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|                                                                                            | Numbers. Pedagogical note: Display all student graphs side-by-side on the board. Ask students: 'Whose graph looks most like a flat horizontal line at the end, and why?' Allow 30-second wait time before cold-calling 4 students.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Students also learned how to construct a basic probability scale and correctly place events (impossible, unlikely, even chance, likely, certain) using real examples from Mama Njeri's stall (e.g., 'probability that a customer buys at least one mandazi = 1').                                                                                                                                                                                                                                                                                                                                                                 | decisions under uncertainty. Pedagogical note: Formally introduce the notation $P(E)$ and ensure all students write it correctly. Common error to address: students dividing favourable outcomes by the number of outcomes in the sample space (theoretical shortcut) instead of by actual trials — clarify that these are two different methods and Lesson 3 will handle theoretical probability.                                                                                                                                                                                                                                                                                                                         |
| <b>Lesson 3</b><br><b>Theoretical Probability and the Sample Space</b>                     | Students listed sample spaces for simple experiments relevant to Mama Njeri's world: tossing a fair coin to decide which flavour of mandazi to make first, rolling a die to simulate customer arrival times, and drawing a mandazi type (plain, coconut, cinnamon) from a bag with equal numbers. Students used tree diagrams and organised lists to enumerate sample spaces systematically, then computed theoretical probabilities by counting equally likely outcomes. Pedagogical note: Stress the phrase 'equally likely outcomes' — this is the foundation of theoretical probability and a frequent source of error when students apply the formula blindly to non-uniform distributions. | Students learned that theoretical probability is defined as $P(E) = (\text{number of favourable outcomes in } E) \div (\text{total number of equally likely outcomes in } S)$ , and that this formula applies only when all outcomes are equally likely. They learned to construct sample spaces using: (1) lists, (2) two-way tables, and (3) simple tree diagrams. Students also learned that the sum of probabilities of all outcomes in a sample space equals 1, and that $P(E') = 1 - P(E)$ , where $E'$ is the complement of event $E$ . Key vocabulary: theoretical probability, equally likely, sample space, complement. | Teacher should explicitly contrast theoretical and experimental probability using the same mandazi scenario: theoretical probability assumes Mama Njeri's bag contains exactly equal numbers of each mandazi type (a controlled model), while experimental probability uses real sales records (messy, real-world data). Connect the complement rule to decision-making: if $P(\text{selling out by noon}) = 0.3$ , then $P(\text{not selling out by noon}) = 0.7$ — Mama Njeri can use this to decide whether to prepare an afternoon batch. Pedagogical note: Have students verify $P(E) + P(E') = 1$ for every event they compute; this self-checking habit prevents arithmetic errors in later, more complex problems. |
| <b>Lesson 4</b><br><b>Mutually Exclusive Events: When Two Things Cannot Happen at Once</b> | Students examined scenarios from Mama Njeri's stall — a customer picks exactly one mandazi type per purchase — and classified pairs of events as mutually exclusive or not. They used Venn diagrams to visualise event overlap, shading $A \cap B$ and noting when the intersection region was empty. Physical card-sort activity: students sorted 12 event-pair cards into 'mutually exclusive' and 'not mutually exclusive' piles, justifying each decision verbally to a partner. Teacher should circulate and listen for correct use of the phrase 'they cannot both happen in the same trial.'                                                                                              | Students learned that two events $A$ and $B$ are mutually exclusive if they cannot both occur in the same trial, formally expressed as $A \cap B = \emptyset$ and therefore $P(A \cap B) = 0$ . Consequently, for mutually exclusive events: $P(A \cup B) = P(A) + P(B)$ . Students also learned that mutually exclusive events are NOT necessarily complementary — complements are always mutually exclusive, but mutually exclusive events need not be complements. Key vocabulary: mutually exclusive, intersection, union, Venn diagram, complement versus mutually exclusive distinction.                                    | Teacher should use the Venn diagram as a persistent visual throughout this lesson and refer back to it during Lesson 6 (Addition Law). Mandazi context: 'A customer cannot simultaneously buy the last plain mandazi AND the last coconut mandazi in a single purchase — these events are mutually exclusive.' Pedagogical note: A critical misconception is that students assume all pairs of events are mutually exclusive. Use cold-calling (minimum 3 students) to test non-obvious cases, e.g., 'Event A: customer spends more than Ksh 20. Event B: customer buys 3 mandazi. Are these mutually exclusive?' Allow 45-second think time before taking responses.                                                      |
| <b>Lesson 5</b><br><b>Independent Events: Does One Outcome</b>                             | Students conducted a two-stage experiment: first tossing a coin (representing Mama Njeri's morning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Students learned that two events $A$ and $B$ are independent if the occurrence of one does not affect the probability of the other,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Teacher should contrast independence with mutual exclusivity using a clear comparison table on the board — these two concepts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| <b>Affect the Next?</b>                                                                                                  | <p>weather — heads = sunny, tails = rainy), then drawing a mandazi type from a bag without any change to the bag's contents. They recorded outcomes in a two-way frequency table and calculated probabilities for each stage, then compared <math>P(\text{mandazi type} \mid \text{sunny})</math> with <math>P(\text{mandazi type} \mid \text{rainy})</math> to check whether the coin outcome changed the draw probability. Result: the probabilities were equal, confirming independence. Pedagogical note: this hands-on test — does the first outcome change the second? — is the conceptual definition students should internalise before the formal algebraic definition is introduced.</p> | <p>formally defined as <math>P(A \cap B) = P(A) \times P(B)</math>. They learned to TEST for independence by checking whether this equation holds, rather than assuming it. Students also learned the informal language: 'knowing A happened gives us no new information about whether B will happen.' Key vocabulary: independent events, dependent events, conditional reasoning (informal), multiplication for independent events.</p>                                                                                                                                                                                                                                                                                                                   | <p>are routinely confused in assessments. Mutually exclusive events CANNOT both happen (<math>P(A \cap B) = 0</math>); independent events CAN both happen and knowing one occurred does not change the other's probability. Mandazi context: Whether it rains in Nairobi this afternoon is independent of whether the coin Mama Njeri flipped this morning landed heads — but whether she sells out is DEPENDENT on whether it rained (fewer customers walk past in heavy rain). Use this dependent example to foreshadow Lesson 7.</p>                                                                                                                                                            |
| <b>Lesson 6</b><br><b>The Addition Law of Probability —</b><br><b>Overlapping Events in</b><br><b>Mama Njeri's World</b> | <p>Students used Venn diagrams to model two overlapping events at Mama Njeri's stall — e.g., Event A: a customer buys plain mandazi; Event B: a customer buys a drink — and physically counted outcomes in each region of the diagram using a class dataset. They noticed that simply adding <math>P(A) + P(B)</math> overcounted customers who bought both, and identified exactly which outcomes were double-counted (the intersection). This discovery moment is pedagogically important — give students 2 minutes to discuss in pairs before teacher confirms the pattern.</p>                                                                                                                | <p>Students learned the Addition Law of Probability: <math>P(A \cup B) = P(A) + P(B) - P(A \cap B)</math>. They learned that subtracting <math>P(A \cap B)</math> corrects for double-counting outcomes that belong to both events. Students recognised the mutually exclusive case from Lesson 4 as the special case where <math>P(A \cap B) = 0</math>, reducing the law to <math>P(A \cup B) = P(A) + P(B)</math>. They practised applying the formula in both directions — finding <math>P(A \cup B)</math> given the other three values, and finding <math>P(A \cap B)</math> given <math>P(A \cup B)</math>, <math>P(A)</math>, and <math>P(B)</math>. Key vocabulary: addition law, union, intersection, double-counting, general addition rule.</p> | <p>Teacher should insist on Venn diagram sketches for every problem — even when students can compute the answer algebraically, the diagram reinforces conceptual understanding and reduces formula misapplication. Pedagogical note: A common exam error is applying <math>P(A \cup B) = P(A) + P(B)</math> without checking mutual exclusivity. Design at least one worked example where students must first determine whether events overlap before choosing which version of the formula to use. Mandazi context: 'What is the probability that a randomly selected customer buys plain mandazi OR buys a drink (or both)?' — directly applicable to Mama Njeri's sales strategy decisions.</p> |
| <b>Lesson 7</b><br><b>Multiplication Law of Probability: Independent and Dependent Events</b>                            | <p>Students conducted two contrasting experiments: (1) Drawing a mandazi type from a bag WITH replacement (independent) and recording the joint outcome of two draws; (2) Drawing WITHOUT replacement (dependent) and recording joint outcomes. They compared the empirical frequencies of joint outcomes across both conditions and noted that the without-replacement results differed from simple multiplication of individual probabilities — setting up the need for a revised formula. Pedagogical note: ensure students record the bag contents before</p>                                                                                                                                 | <p>Students learned that: (1) For INDEPENDENT events, <math>P(A \cap B) = P(A) \times P(B)</math>; (2) For DEPENDENT events (sampling without replacement), <math>P(A \cap B) = P(A) \times P(B A)</math>, where <math>P(B A)</math> is the conditional probability of B given A has already occurred — computed by adjusting the sample space after the first draw. Students learned to identify which formula applies by asking: 'Does the first outcome change the conditions for the second?' Key vocabulary: multiplication law, with replacement, without replacement, conditional probability (informal), dependent</p>                                                                                                                              | <p>Teacher should use explicit fraction arithmetic for the without-replacement case, showing how both the numerator (favourable outcomes) and denominator (total outcomes) change after the first draw. Mandazi context: Mama Njeri has 5 plain and 3 coconut mandazi left. A customer picks two at random without looking — what is the probability both are plain? Work through this step-by-step: <math>P(\text{first plain}) = 5/8</math>; <math>P(\text{second plain} \mid \text{first was plain}) = 4/7</math>; <math>P(\text{both plain}) = 5/8 \times 4/7 = 20/56 = 5/14</math>. Pedagogical note: This is students' first informal encounter with conditional</p>                         |

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|                                                                                  | and after each draw in the dependent condition; this makes the changing probability visible and concrete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | probability — do not use the notation $P(B A)$ formally; instead use the language 'given that A already happened, what is the new probability of B?'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Lesson 8</b><br><b>Probability Tree Diagrams: Mapping Sequential Outcomes</b> | Students were given a partially completed probability tree diagram representing two sequential mandazi draws (with replacement) and were asked to complete the branches, label each branch with its probability, and calculate endpoint probabilities by multiplying along branches. They then constructed a full tree diagram from scratch for a three-stage scenario: weather (sunny/rainy), customer arrival (early/late), and mandazi type purchased (plain/coconut). Pedagogical note: Insist that students (1) label every branch with a probability, (2) verify that branches from each node sum to 1, and (3) verify that all endpoint probabilities sum to 1. These three checks catch almost all tree diagram errors.                               | Students learned that a probability tree diagram is a visual tool for systematically mapping all possible outcomes of a sequential (multi-stage) experiment, where each branch represents one possible outcome of one stage, branch labels show the probability of that outcome, and endpoint probabilities are found by multiplying probabilities along the path (Multiplication Law). Students also learned to use tree diagrams to find $P(A \cup B)$ for sequential events by adding the relevant endpoint probabilities (Addition Law). Key vocabulary: tree diagram, branch, stage, endpoint, path probability, systematic enumeration.                                                                                                                                  | Teacher should emphasise that tree diagrams are a problem-solving tool — they make invisible structure visible — not merely a drawing exercise. Mandazi context: Mama Njeri wants to know: 'What is the probability that on two consecutive market days, she sells out at least once?' A tree diagram with outcomes (sell out / not sell out) on each branch, labelled with experimental probabilities from her records, answers this directly. Pedagogical note: For the 'at least one' probability, teach the complement shortcut: $P(\text{at least one sell-out}) = 1 - P(\text{no sell-out on either day})$ . This reinforces the complement rule from Lesson 3 and demonstrates how lessons connect across the unit.                                                                                                                                                                                                                    |
| <b>Lesson 9</b><br><b>Synthesis &amp; Unit Assessment: Helping Mama Njeri</b>    | Students worked through a multi-part performance task set entirely in Mama Njeri's mandazi stall context. Given a dataset of 60 days of sales records, students: (1) computed experimental probabilities for each mandazi type, (2) constructed a sample space for a two-item customer order, (3) identified mutually exclusive and independent event pairs, (4) applied the Addition Law to find the probability a customer buys plain OR coconut mandazi, (5) applied the Multiplication Law (with and without replacement), and (6) drew a tree diagram for a two-day sequential sales scenario. Teacher should observe whether students select the correct tool for each sub-problem — this metacognitive skill is the highest-level outcome of the unit. | Students synthesised all unit concepts: experimental probability ( $P(E) = \text{favourable trials} \div \text{total trials}$ ), theoretical probability ( $P(E) = \text{favourable outcomes} \div \text{sample space size}$ , equally likely outcomes), the complement rule ( $P(E') = 1 - P(E)$ ), mutually exclusive events ( $P(A \cap B) = 0$ ; $P(A \cup B) = P(A) + P(B)$ ), independent events ( $P(A \cap B) = P(A) \times P(B)$ ), the Addition Law ( $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ ), the Multiplication Law for independent and dependent events, and probability tree diagrams for sequential experiments. Students also reflected on how probability reasoning supports real decision-making under uncertainty — the driving question's core answer. | Teacher should facilitate a whole-class debrief after the performance task, returning explicitly to the driving question: 'How can mathematics help Mama Njeri make smarter decisions even when she cannot be certain of the outcome?' Draw a concept map on the board connecting all nine lessons' key ideas, showing how experimental probability (Lessons 1–2) built intuition, theoretical probability (Lesson 3) provided a precise model, event relationships (Lessons 4–5) refined that model, the Addition and Multiplication Laws (Lessons 6–7) gave computational tools, and tree diagrams (Lesson 8) provided a unifying visual framework. Pedagogical note: Use this lesson to address any persistent misconceptions surfaced by the assessment — common ones include: applying the simple addition rule to non-mutually-exclusive events, multiplying without checking for dependence, and misidentifying the sample space size. |